Milo

Fixed-price orders with authorized card payments and a Midnight chain record. Project status, before and after.

Test mode and Midnight Preprod only. No real money has moved.

Date	26 October 2026
Last commit	daaf401 on main at github.com/symulacr/milo, pushed 26 October 2026. Everything described here is committed.
Live site	https://milo-nine-delta.vercel.app (Vercel), Convex dev deployment
Checks	biome, tsc, strict tsc and build pass. 1,368 unit tests pass, 0 fail. Taste lint 0 failures.

Summary

What changed, what is still open, and what the project needs from its owner.

+521
unit tests (811 to 1,332)

16 to 0
taste lint failures

7 to 14
of 14 contract circuits reachable from the UI

195 to 6
queries to list 64 orders

Where the project stands

- The prototype became a live product: magic-link and wallet sign-in, self-serve orders in Convex, a Lace wallet flow, and a server-side test-card hold.
- An 80-finding code audit was run. About 59 findings are fixed with tests. The rest are open or deliberately deferred.
- The full order path has never been run end to end on Preprod with a real Lace wallet and a Stripe test capture. That is the main open risk.
- Magic-link email is sent through AgentMail from mi.lo@agentmail.to and reaches any recipient. The free plan allows 100 sends a day and 3 new recipients in the first hour, 5 in the first day and 10 in the first week.

What the project does not claim

- It is not an escrow service. Stripe says it does not provide escrow and lists escrow services as restricted. Milo authorizes a card at order time and captures it on acceptance.
- It is not live. Stripe is in test mode and the chain is Midnight Preprod.
- Nothing is audited by a third party.

What Milo is

One buyer, one studio, one fixed price. Terms stay off-chain. The order phase and commitments are recorded on Midnight.

Layer	What it does	Technology
Web app	Landing, sign-in, quotes, orders, account, legal pages	React 19, react-router, Tailwind 4, shadcn
Backend	Orders, invites, payments state, webhooks, access rules, email	Convex, Better Auth, AgentMail (Resend and Brevo supported)
Payments	Card authorized at order time, captured on acceptance, released on expiry or cancel	Stripe manual capture (test mode)
Chain	14-circuit order contract: phase, deadlines, dispute, resolution	Midnight Compact, Lace wallet, Preprod
Hosting	Static site with security headers and a CSP in report-only mode	Vercel

Order timeline

Four windows, each counted from the previous step: 1 hour to accept, 24 hours to deliver, 48 hours to review, 48 hours to resolve a dispute. Together they stay inside the 7-day card authorization window (121 hours).

Before and after

Measured during this project. Rows marked estimate were not re-measured.

Measure	Before	After
Web source lines, non-test	6,738	6,286
app.css lines	361	140
Unit tests passing	811	1,368
Duplicate-code clones (estimate)	86	about 66
Contract circuits with UI	7 of 14	14 of 14
Lace deploy transactions	3	2
Quote form fields	5	3
Words on main app screens	452	304
Landing words	496	274
Landing transfer	152 KB	111 KB
JavaScript on the landing page	36 KB	0.8 KB
First paint on the landing page	368 ms	136 ms
Queries to list 64 orders	195	6
Wasm compile on a repeat visit	80 ms	0.1 ms (cache hit)
Contract test circuit runs	177	46
Contract compile, unchanged source	166 s	2.3 s
Quality gate, nothing changed	27 s	0.07 s
Chain code coverage, functions / lines	74.7% / 78.2%	91.4% / 93.4%
Access test (functions by user type)	none	126 rows

Targets of 4x faster shipping, 4x fewer words and a 4x smaller landing page were not reached. The measured gains were 1.5x cold shipping, about 1.3x to 1.8x fewer words, and 1.4x smaller landing transfer.

Security posture

Controls in place, their evidence, and what remains.

Control	State	Residual risk
Access to data	A 126-row test covers every public function against anonymous, stranger, buyer, merchant, operator and revoked users	New functions need a row; the test fails if one is missing
Invite links	Token rotates when the order is claimed; removed from the address bar; reads need the invited account	Lifetime of unclaimed tokens not re-verified
Payments	Idempotency keys on capture and void; webhook signature and replay checks; refund, dispute and expiry events recorded; live keys refused	Live mode, Connect and payout are not built
Browser secrets	Encrypted with a non-extractable key; exports encrypted	Does not stop XSS or a malicious extension
Wasm integrity	SHA-256 recorded at build and checked at load	Hash is from the same build
Headers	HSTS, nosniff, frame deny, referrer policy; CSP in report-only mode with a live report receiver	CSP not yet enforcing
Sign-in	Magic links and wallet signatures, both rate limited; single-use tokens	Free email plan caps new recipients; a custom domain removes the cap
Contract	Role-gated circuits, deadlines, immutable dispute evidence	An escalated dispute cannot be resolved (C-04). Fix is written and held for a redeploy.

Audit result

Four workers read the code in parallel and each finding was tied to a file, a line and evidence.

Severity	Found	State
Critical	4	3 fixed (payment recovery, and the work is now committed). 1 open: an escalated dispute cannot resolve (contract redeploy held).
High	22	Fixed with tests, apart from the key-storage item, which is now partly addressed
Medium	32	Most fixed. CSP enforcing and a few smaller items remain.
Low	12	Mostly fixed
Warning	10	Noted. No action required for most.

Two audit items were wrong and were corrected: an unused rate-limit code was not a defect, and two top payment risks reported later did not exist (capture already sends an idempotency key, and the capture deadline is already stored).

Open items and what unlocks them

Owner action	What it unlocks
Buy a domain and verify it with an email provider	Removes the new-recipient cap and gives the site a custom domain
Send operator emails	Someone can resolve disputes
Rotate the AgentMail key	The key was shared in a chat message
Run one order with Lace, tDUST and a Stripe test card	The first end-to-end proof and real CSP reports to enforce from
Approve the contract redeploy	Disputes on unreviewed deliveries become resolvable
Decide the payout model and the word escrow	Final legal wording, and whether Stripe Connect is needed
Move Vercel to Pro	Commercial use. The free plan forbids processing payments.

Roadmap

1. Done: committed and pushed to GitHub, with architecture notes, decision records and an audit register. Next: a CI build that deploys the prebuilt site to Vercel, because the contract must be compiled first.
2. Domain, email and operator setup.
3. One full Preprod order, then enforce the CSP.
4. Redeploy the fixed contract and migrate open orders.
5. Decide live Stripe mode, a production Convex deployment and backups.

Rewrite or continue

Research recommends continuing. The stack is already current, and a rewrite would discard the 14-circuit contract, the order machine and 17,000 lines of tests while rebuilding the same missing end-to-end path. Estimated cost is about 15 engineer-weeks to consolidate, against 24 to 32 for a rewrite. Several of those consolidation stages are already done.

How the numbers were measured: line counts from the working tree on 26 October 2026; test counts from the repository unit test run; paint and transfer from headless Chromium runs; query counts from an instrumented database double in tests. Estimates are marked.